

Ethnography

- Ethnography is a method for studying users in their natural surroundings.
- Representing the information gleaned from an ethnographic study so that it can be used in design has been problematic.
- The goals of ethnography are to study the details, while the goals of system design are to produce abstractions; hence they are not immediately compatible.

Ethnographic Techniques

- Ethnographic techniques are various techniques and methods adopted to gather data while studying people.
 - Observing Users
 - Asking Users
 - Interviews
 - Questionnaires
 - Asking Experts
 - Inspections
 - Walkthroughs

Observing Users ...

- Observation involves watching and listening to users.
- Observing users interacting with software, even casual observing, can tell you an enormous amount about what they do, the context in which they do it, how well technology supports them, and what other support is needed.
- Users can be observed in controlled laboratory-like conditions, as in usability testing, or in the natural environments in which the products are used-i.e., the field.
- How the observation is done depends on why it is being done and the approach adopted.
- There is a variety of structured, less structured, and descriptive observation techniques for evaluators to choose from.
- Which they select and how their findings are interpreted will depend upon the evaluation goals, the specific questions being addressed, and practical constraints.

What and when to observe

Table 12.1 Type of observation

Observation	Controlled environment (i.e., lab-like)	Field environment (i.e., natural)
Outsider looking on	"Quick and dirty" In usability testing	"Quick and dirty" In field studies
Insider	(Not applicable)	Participant observation (e.g., in ethnography)

Approaches to observation

- Observers can be outsiders in the field and in the controlled environments, but they can't be insiders in a controlled environment.
- In the field it is possible to have varying degrees of "insider outsider ness." In practice these distinctions are more difficult to describe than to experience!

"Quick and dirty" observation

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- In the field it is possible to have varying degrees of "insider outsider ness." In practice these distinctions are more difficult to describe than to experience!
- "Quick and dirty" observations can occur anywhere, anytime.
- For example, evaluators often go into a school, home, or office to watch and talk to users in a casual way to get immediate feedback about a prototype or product.
- Evaluators can also join a group for a short time, which gives them a slightly more insider role.
- Quick and dirty observations are just that, ways of finding out what is happening quickly and with little formality.

Observation in usability testing

- Video and interaction logs capture everything that the user does during a usability test including keystrokes, mouse clicks, and their conversations.
- In addition, observers can watch through a one-way mirror or via a remote TV screen.
- The observational data is used to see and analyze what users do and how long they spend on different aspects of the task.
- It also provides insights into users' affective reactions. For example, sighs, tense shoulders, frowns, and scowls speak of users' dissatisfaction and frustrations.
- The environment is controlled but users often forget that they are being observed.
- In addition, many evaluators also supplement findings from the laboratory with observations in the field.

Observation in field studies...

- In field studies, as we have said, observers may be anywhere along the outsider- insider spectrum.
- Looking on as an outsider, being a participant observer, or being an ethnographer brings a philosophy and practices that influence what data is collected, how data collection is done, and how the data is analyzed and reported.
- Colin Robson (1993) summarizes the possible levels of participation as: complete participants, more marginal participants, observers who also participate, and people who observe from the outside and do not participate.
- Whether and in what ways observers influence those being observed depends on the type of observation and the observer's skills.
- The goal is to cause as little disruption as possible. An example of outsider observation is when an observer is interested only in the presence of certain types of behavior.

...Observation in field studies

- Inside observers may be participant observers or ethnographers.
- In participant observation evaluators participate with users in order to learn what they do and how and why they do it.
- A fully participant observer observes from the inside as a member of the group, which means she must not only be present to share experiences, but also learn the social conventions of the group, including beliefs and protocols, dress codes, communication conventions, use of language, and non-verbal communication.
- "Participant observation combines participation in the lives of the people under study with maintenance of a professional distance that allows adequate observation and recording of data" (Fetterman, 1998, p. 34-35).

Asking Users – Interviews...

- Interviews are used in "quick and dirty" evaluation, in usability testing, and in field studies to ask about facts, behavior, beliefs, and attitudes.
- Interviews can be thought of as a "conversation with a purpose" (Kahn and Cannell, 1957).
- How like an ordinary conversation the interview is depends on the questions to be answered and the type of interview method used.
- There are four main types of interviews:
 - open-ended or unstructured,
 - structured,
 - semi-structured,
 - and group interviews
- The first three types are named according to how much control the interviewer imposes on the conversation by following a predetermined set of questions.
- The fourth involves a small group guided by an interviewer who facilitates discussion of a specified set of topics.
- The most appropriate approach to interviewing depends on the evaluation goals, the questions to be addressed, and the paradigm adopted.

...Asking Users – Interviews...

Unstructured interviews

- Open-ended or unstructured interviews are at one end of a spectrum of how much control the interviewer has on the process.
- They are more like conversations that focus on a particular topic and may often go into considerable depth.
- Questions posed by the interviewer are open, meaning that the format and content of answers is not predetermined.
- The interviewee is free to answer as fully or as briefly as she wishes. Both interviewer and interviewee can steer the interview.
- Thus one of the skills necessary for this type of interviewing is to make sure that answers to relevant questions are obtained. It is therefore advisable to be organized and have a plan of the main things to be covered.
- Going in without an agenda to accomplish a goal is not advisable, and should not to be confused with being open to new information and ideas.
- A benefit of unstructured interviews is that they generate rich data. Interviewees often mention things that the interviewer may not have considered and can be further explored.
- But this benefit often comes at a cost. A lot of unstructured data is generated, which can be very time-consuming and difficult to analyze.
- It is also impossible to replicate the process, since each interview takes on its own format.

...Asking Users – Interviews...

Structured interviews

- Structured interviews pose predetermined questions similar to those in a questionnaire
- Structured interviews are useful when the study's goals are clearly understood and specific questions can be identified.
- To work best, the questions need to be short and clearly worded., Responses may involve selecting from a set of options that are read aloud or presented on paper.
- The questions should be refined by asking another evaluator to review them and by running a small pilot study.
- Typically the questions are closed, which means that they require a precise answer.
- The same questions are used with each participant so the study is standardized.

...Asking Users – Interviews...

Semi-Structured interviews

- Semi-structured interviews combine features of structured and unstructured interviews and use both closed and open questions.
- For consistency the interviewer has a basic script for guidance, so that the same topics are covered with each interviewee.
- The interviewer starts with preplanned questions and then probes the interviewee to say more until no new relevant information is forthcoming.

...Asking Users – Interviews

Group interviews

- One form of group interview is the focus group that is frequently used in marketing, political campaigning, and social sciences research.
- Normally three to 10 people are involved. Participants are selected to provide a representative sample of typical users; they normally share certain characteristics.
- For example, in an evaluation of a university website, a group of administrators, faculty, and students may be called to form three separate focus groups because they use the web for different purposes.
- The benefit of a focus group is that it allows diverse or sensitive issues to be raised that would otherwise be missed.
- The method assumes that individuals develop opinions within a social context by talking with others.
- Often questions posed to focus groups seem deceptively simple but the idea is to enable people to put forward their own opinions in a supportive environment.
- A preset agenda is developed to guide the discussion but there is sufficient flexibility for a facilitator to follow unanticipated issues as they are raised.
- The facilitator guides and prompts discussion and skillfully encourages quiet people to participate and stops verbose ones from dominating the discussion.
- The discussion is usually recorded for later analysis in which participants may be invited to explain their comments more fully.

Asking users - Questionnaires

- Questionnaires are a well-established technique for collecting demographic data and users' opinions.
- They are similar to interviews and can have closed or open questions.
- Effort and skill are needed to ensure that questions are clearly worded and the data collected can be analyzed efficiently.
- Questionnaires can be used on their own or in conjunction with other methods to clarify or deepen understanding.
- The methods and questions used depends on the context, interviewees and so on.
- The questions asked in a questionnaire, and those used in a structured inter- view are similar, so how do you know when to use which technique?
- One advantage of questionnaires is that they can be distributed to a large number of people. Used in this way, they provide evidence of wide general opinion.
- On the other hand, structured interviews are easy and quick to conduct in situations in which people will not stop to complete a questionnaire.

Asking Experts - Inspections

- Sometimes users are not easily accessible or involving them is too expensive or takes too long.
- In such circumstances, experts or combinations of experts and users can provide feedback.
- Various inspection techniques began to be developed as alternatives to usability testing in the early 1990s.
- These included various kinds of expert evaluations or reviews, such as heuristic evaluations and walkthroughs, in which experts inspect the human-computer interface and predict problems users would have when interacting with it.
- Typically these techniques are relatively inexpensive and easy to learn as well as being effective, which makes them appealing.
- They are similar to some software engineering practices where code and other types of inspections have been conducted for years.
- In addition, they can be used at any stage of a design project, including early design before well-developed prototypes are available.

...Asking Experts - Inspections

Heuristic evaluation

- Heuristic evaluation is an informal usability inspection technique developed by Jakob Nielsen and his colleagues (Nielsen, 1994a) in which experts, guided by a set of usability principles known as heuristics, evaluate whether user-interface elements, such as dialog boxes, menus, navigation structure, online help, etc., conform to the principles.
- These heuristics closely resemble the high-level design principles and guidelines discussed in Chapters 1 and 8, e.g., making designs consistent, reducing memory load, and using terms that users understand. When used in evaluation, they are called heuristics.
- The original set of heuristics was derived empirically from an analysis of 249 usability problems (Nielsen, 1994b).
- We list the latest here (also in Chapter I), this time expanding them to include some of the questions addressed when doing evaluation:

...Asking Experts - Inspections

Heuristic evaluation

- Visibility of system status Are users kept informed about what is going on? Is appropriate feedback provided within reasonable time about a user's action?
- Match between system and the real world Is the language used at the interface simple? Are the words, phrases and concepts used familiar to the user?
- User control and freedom Are there ways of allowing users to easily escape from places they unexpectedly find themselves in?
- Consistency and standards Are the ways of performing similar actions consistent? Help users recognize, diagnose, and recover from errors Are error messages helpful? Do they use plain language to describe the nature of the problem and suggest a way of solving it?
- Error prevention Is it easy to make errors? If so where and why? Recognition rather than recall Are objects, actions and options always visible?
- Flexibility and efficiency of use Have accelerators (i.e., shortcuts) been provided that allow more experienced users to carry out tasks more quickly?
- Aesthetic and minimalist design Is any unnecessary and irrelevant information provided?
- Help and documentation Is help information provided that can be easily searched and easily followed?

Asking Experts - Walkthroughs

- Walkthroughs are an alternative approach to heuristic evaluation for predicting users' problems without doing user testing.
- As the name suggests, they involve walking through a task with the system and noting problematic usability features.
- Most walkthrough techniques do not involve users.
- Others, such as pluralistic walkthroughs, involve a team that includes users, developers, and usability specialists.

Asking Experts - Walkthroughs

Cognitive walkthroughs

- "Cognitive walkthroughs involve simulating a user's problem-solving process at each step in the human-computer dialog, checking to see if the user's goals and memory for actions can be assumed to lead to the next correct action." (Nielsen and Mack, 1994, p. 6).
- The defining feature is that they focus on evaluating designs for ease of learning—a focus that is motivated by observations that users learn by exploration (Wharton et al., 1994). The steps involved in cognitive walkthroughs are:

Asking Experts - Walkthroughs

Cognitive walkthroughs

- The characteristics of typical users are identified and documented and sample tasks are developed that focus on the aspects of the design to be evaluated. A description or prototype of the interface to be developed is also produced, along with a clear sequence of the actions needed for the users to complete the task.
- A designer and one or more expert evaluators then come together to do the analysis.
- The evaluators walk through the action sequences for each task, placing it within the context of a typical scenario, and as they do this they try to answer the following questions: Will the correct action be sufficiently evident to the user? (Will the user know what to do to achieve the task?) Will the user notice that the correct action is available? (Can users see the button or menu item that they should use for the next action? Is it apparent when it is needed?) Will the user associate and interpret the response from the action correctly? (Will users know from the feedback that they have made a correct or incorrect choice of action?) In other words: will users know what to do, see how to do it, and understand from feedback whether the action was correct or not?
- As the walkthrough is being done, a record of critical information is compiled in which: The assumptions about what would cause problems and why are recorded. This involves explaining why users would face difficulties, Notes about side issues and design changes are made. A summary of the results is compiled.
- The design is then revised to fix the problems presented. It is important to document the cognitive walkthrough, keeping account of what works and what doesn't.

Asking Experts - Walkthroughs

Pluralistic walkthroughs

- "Pluralistic walkthroughs are another type of walkthrough in which users, developers and usability experts work together to step through a [task] scenario, discussing usability issues associated with dialog elements involved in the scenario steps" (Nielsen and Mack, 1994, p. 5).
- Each group of experts is asked to assume the role of typical users. The walkthroughs are then done by following a sequence of steps (Bias, 1994):

Asking Experts - Walkthroughs

Pluralistic walkthroughs

- Scenarios are developed in the form of a series of hard-copy screens representing a single path through the interface. Often just two or a few screens are developed.
- The scenarios are presented to the panel of evaluators and the panelists are asked to write down the sequence of actions they would take to move from one screen to another. They do this individually without conferring with one another.
- When everyone has written down their actions, the panelists discuss the actions that they suggested for that round of the review. Usually, the representative users go first so that they are not influenced by the other panel members and are not deterred from speaking. Then the usability experts present their findings, and finally the developers offer their comments.
- Then the panel moves on to the next round of screens. This process continues until all the scenarios have been evaluated.
- The benefits of pluralistic walkthroughs include a strong focus on users' tasks. Performance data is produced and many designers like the apparent clarity of work-ing with quantitative data.
- The approach also lends itself well to participatory design practices by involving a multidisciplinary team in which users play a key role.
- Limitations include having to get all the experts together at once and then proceed at the rate of the slowest. Furthermore, only a limited number of scenarios, and hence paths through the interface, can usually be explored because of time constraints. .